

PostgreSQL Migration Guide

Environment Variables

- DJANGO_DB_ENGINE=postgres
- POSTGRES_DB, POSTGRES_USER, POSTGRES_PASSWORD, POSTGRES_HOST, POSTGRES_PORT configure production database.
- No code changes are required to switch from SQLite to PostgreSQL.

Setup Steps

- Install PostgreSQL 15/16.
- Create database and user.
- Set .env values.
- Run python manage.py migrate.
- Run python manage.py create_demo_accounts if demo/starter users are required.
- Run python manage.py collectstatic before production service start.

Data Migration

- Back up SQLite with python manage.py backup_data.
- Switch environment to PostgreSQL and run migrations.
- Load backup JSON only after verifying target database is correct.
- Validate dashboards, reports and patient lookup after import.

Production Tuning

- Use SSD/NVMe storage.
- Tune shared_buffers, work_mem, maintenance_work_mem and max_connections.
- Enable regular pg_dump backups.
- Monitor slow queries and disk growth.